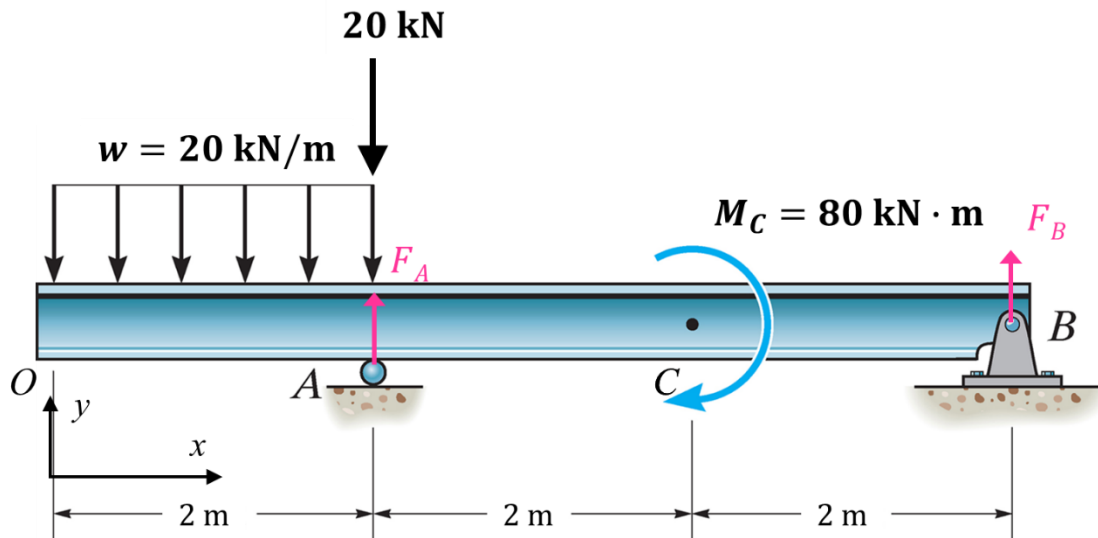


1. Determine the absolute maximum normal stress (in unit of MPa) in the beam with external loadings shown below. The beam has a uniform square cross-section with lateral size $a = 0.2$ m. Note: (1) the shear and moment diagrams can be calculated by either section method or graphical method; (2) there is a concentrated load at point A and a bending moment at point C . (15' in total)



Solutions:

Graphical method to calculate the shear and moment diagrams.

Step 1: Support Reactions

Applying free-body diagram to the entire beam, the force equation of equilibrium gives:

$$\sum F_y = 0 \rightarrow$$

$$F_A + F_B = 20 \times 2 + 20 = 60 \text{ kN}$$

$$\sum M_A = 0 \rightarrow$$

$$F_B \times 4 + 20 \times 2 \times 1 = 80 \text{ kN} \cdot \text{m}$$

Then,

$$F_A = 50 \text{ kN}$$

$$F_B = 10 \text{ kN}$$

Step 2: Shear Diagram

a) Determine the shear at the left end:

$$x = 0, V = 0 \text{ kN}$$

b) Determine **V** diagram according to $\frac{dV}{dx} = w$ for OA region:

$$\frac{dV}{dx} = w = -20$$

$$\rightarrow V = -20x + C_1$$

The constant C_1 can be determined by applying the “boundary” condition ($x = 0$, $V = 0$ kN). Thus, $C_1 = 0$ and $V = -20x$ ($0 \leq x \leq 2$).

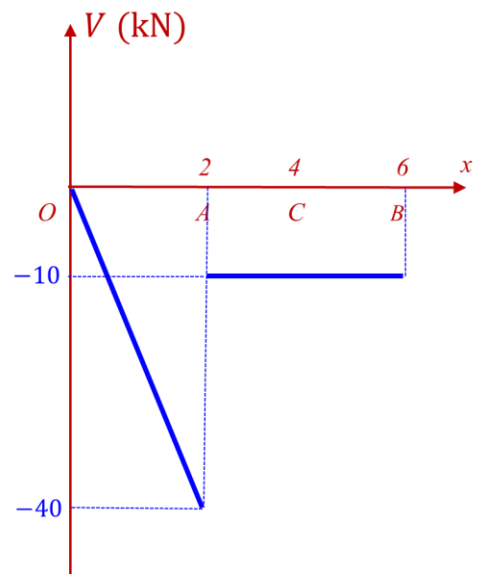
c) At point *A* ($x = 2$ m), due to the upward support force $F_A = 50$ kN and the downward concentrated force 20 kN, the shear force **V** has an upward jump by 30 kN, i.e. **V** jumps from -40 kN to -10 kN.

d) Determine **V** diagram according to $\frac{dV}{dx} = w$ for AB region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow V = C_2$$

The constant C_2 can be determined by the “continuity” condition at the last point of *OA* region after jump ($x = 2$ m, $V = -10$ kN). Thus, $C_2 = -10$ kN and $V = -10$ kN ($2 \text{ m} \leq x \leq 6 \text{ m}$).



Step 3: Moment Diagram

a) Determine **M** at left end first (the origin *A* of *x*-axis):

$$x = 0, M = 0$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for the OA region:

$$\frac{dM}{dx} = V = -20x$$

$$\rightarrow M = -10x^2 + C_3$$

The constant C_3 can be determined by applying the “boundary” condition ($x = 0$, $M = 0$). Thus, $C_3 = 0$ and $M = -10x^2$ ($0 \leq x \leq 2$ m).

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for AC region:

$$\frac{dM}{dx} = V = -10 \text{ kN}$$

$$\rightarrow M = -10x + C_4$$

The constant C_4 can be determined by applying the “continuity” condition at last point A of the AD region ($x = 2$, $M = -40 \text{ kN} \cdot \text{m}$). Thus, $C_4 = -20$ and $M = -10x - 20$ ($2 \text{ m} \leq x \leq 4 \text{ m}$).

d) At point C ($x = 4 \text{ m}$), due to the clockwise external bending moment $M_C = 80 \text{ kN} \cdot \text{m}$, the moment $\mathbf{M}$ has an upward jump by $M_C = 80 \text{ kN} \cdot \text{m}$, i.e. $\mathbf{M}$ jumps from $-60 \text{ kN} \cdot \text{m}$ to $+20 \text{ kN} \cdot \text{m}$.

e) Determine $\mathbf{M}$ diagram according to $\frac{dM}{dx} = V$ for $\underline{CB}$ region:

$$\frac{dM}{dx} = V = -10 \text{ kN}$$

$$\rightarrow M = -10x + C_5$$

The constant C_5 can be determined by applying the “continuity” condition at last point C after jump ($x = 4 \text{ m}$, $M = +20 \text{ kN} \cdot \text{m}$). Thus, $C_5 = 60 \text{ kN} \cdot \text{m}$ and $M = -10x + 60$ ($4 \text{ m} \leq x \leq 6 \text{ m}$).

Therefore, the maximum moment throughout the entire beam is $M_{\max} = -60 \text{ kN} \cdot \text{m}$ at point C ($x = 4 \text{ m}$).

Step 4: Section Property

The moment of inertia I of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.2 \times 0.2^3 = 1.3333 \times 10^{-4} \text{ m}^4$$

Step 4: Maximum Normal Stress

The absolute maximum normal stress is determined by:

$$\sigma_{\max} = \frac{M_{\max}c}{I}$$

$$\text{Thus, } \sigma_{\max} = \frac{60000 \times 0.1}{1.3333 \times 10^{-4}} = \boxed{45 \text{ MPa}}$$

